

Dummy RAPM

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Abstract

Traditional Adjusted Plus-Minus (APM) models regress the home team point differential on a matrix of player indicator variables determining who was on the court during that stint. Typically, low-minute players are excluded from the regression due to their unstable coefficients. When a player is excluded, their column is removed from the design matrix, and their effect is implicitly absorbed into the coefficients of the remaining players who shared the court with them. This induces a systematic bias in the coefficient estimates of retained players whose court time overlaps heavily or rarely with excluded players, depending on their role in the team.

We introduce Dummy RAPM, an incremental improvement to RAPM which explicitly includes the count of excluded low-minute players on the court for each team as an additional regressor. Rather than allowing excluded players' contributions to be silently redistributed across the design matrix, this approach isolates their effect into 5 coefficients (representing 5 possible numbers of low-minute players in a single lineup), preventing it from contaminating the estimates of retained players. We find that this reallocation yields improved out-of-sample predictive performance than RAPM.

1 Introduction

Plus-minus is a basketball statistic that measures the net points scored while a player is on the court. In its basic form it fails to account for the quality of teammates and opponents, so [Rosenbaum \(2004\)](#) introduced Adjusted Plus-Minus (APM), which addresses this by regressing stint-level point differentials on player indicators, isolating each player’s individual contribution.

A limitation of APM is that sparse playing time leads to unstable estimates. [Sill \(2010\)](#) tackled this by applying ridge regression, producing Regularized APM (RAPM), which shrinks unreliable parameters towards zero rather than relying on Ordinary Least Squares estimates. Bayesian methods have also been explored as an alternative means of stabilizing coefficients while incorporating prior basketball knowledge. [Deshpande and Jensen \(2016\)](#) extended this framework further by regressing on change in win probability rather than point differential, more directly capturing each player’s impact on game outcomes.

2 Traditional APM Model Structure

A player’s APM is estimated via a weighted linear regression model in which each observation corresponds to a single stint — a period of play with no lineup change. The dependent variable $\mathbf{y}$ is a vector of net points for the home team per possession scored during that stint:

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad y_i \in \mathbb{R}$$

where each observation is weighted by the number of possessions in that stint, so that longer stints contribute more to the estimation. The design matrix $\mathbf{X}$ encodes which players are on the court for each stint, where each entry takes the value 1 if the player is on the home team, -1 if on the away team, and 0 if not on the court:

$$\mathbf{X} = \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,p} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,p} \end{pmatrix}, \quad x_{i,j} \in \{-1, 0, 1\}$$

The regression is then estimated as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where $\boldsymbol{\beta}$ is a vector of player coefficients, each representing that player’s average point contribution per possession relative to their opponents, adjusting for the other players on the court and home/away effects.

3 Ridge Regression Model

Ridge Regression models have been introduced in order to improve predictive performance by shrinking coefficients towards zero, which is particularly beneficial in the context of APM models. The ternary nature of the entries $x_{i,j} \in \{-1, 0, 1\}$ can induce strong correlations between columns of $\mathbf{X}$, inflating the variance of $\hat{\boldsymbol{\beta}}_{\text{OLS}}$. Ridge shrinkage reduces this variance at the cost of a small bias, yielding a lower mean squared error overall:

$$\text{MSE}(\hat{\boldsymbol{\beta}}_{\text{ridge}}) = \underbrace{\text{Bias}^2}_{\text{increases with } \lambda} + \underbrace{\text{Var}}_{\text{decreases with } \lambda}$$

Furthermore, many players appear in relatively few stints, meaning their individual signal is weak and OLS will overfit to noise in those small samples. Shrinkage pulls low-minute players toward zero while allowing high-minute players’ coefficients to reflect their true impact.

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = \arg \min_{\boldsymbol{\beta}} \{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|^2 \}$$

4 Typical Filtering Process

As mentioned above, ridge regression partially mitigates the low per-player sample problem by shrinking uncertain coefficients toward zero. However, it does not eliminate it entirely, a player appearing in only a handful of stints will still receive an unreliable estimate, merely one that is attenuated rather than unconstrained. Filtering therefore remains standard practice even when regularisation is applied: players with insufficient playing time are excluded from the design matrix $\mathbf{X}$ prior to estimation, with a common threshold being any player who has participated in fewer than some minimum number of stints k across the sample (or number of minutes played during the season).

When a player i is excluded, their column $x_{\cdot,i}$ is removed from $\mathbf{X}$, reducing the dimensionality of the problem. The stints in which an excluded player appeared are retained in $\mathbf{y}$, but their contribution is implicitly absorbed into the coefficients of the remaining players who were on the court during those stints, potentially biasing those estimates.

5 Dummy Variable Methodology

Rather than allowing the excluded players' contributions to be absorbed by their teammates, an alternative approach is to explicitly account for them via a set of dummy columns. For each stint j , we define five dummy variables $d_j^{(k)}$ for $k = 1, \dots, 5$, where $d_j^{(k)}$ takes the value 1 if exactly k excluded players are on the court during possession j , and 0 otherwise:

$$d_j^{(k)} = \begin{cases} 1 & \text{if exactly } k \text{ excluded players are on the court during stint } j \\ 0 & \text{otherwise} \end{cases}$$

This yields an augmented design matrix $\mathbf{X}^*$ in which the five dummy columns replace all excluded player columns:

$$\mathbf{X}^* = \begin{pmatrix} x_{1,1} & \cdots & x_{1,p^*} & d_1^{(1)} & d_1^{(2)} & d_1^{(3)} & d_1^{(4)} & d_1^{(5)} \\ x_{2,1} & \cdots & x_{2,p^*} & d_2^{(1)} & d_2^{(2)} & d_2^{(3)} & d_2^{(4)} & d_2^{(5)} \\ \vdots & & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & \cdots & x_{n,p^*} & d_n^{(1)} & d_n^{(2)} & d_n^{(3)} & d_n^{(4)} & d_n^{(5)} \end{pmatrix}$$

where p^* denotes the number of retained players after filtering. The coefficient $\hat{\beta}^{(k)}$ on the k -th dummy can be interpreted as the average net point contribution per possession when exactly k excluded players are on the court, allowing the model to account for varying degrees of scrub presence without having to estimate individual player coefficients.

6 Defining a Scrub

The "reference players" in [Rosenbaum \(2004\)](#)'s methodology were defined as a player that had played at least 250 minutes across two seasons. Considering that an NBA season has 82 games, this equates to someone that played less than roughly 1.5 minutes per game across his two season study period.

Instead of using a total minute or stint threshold across the training set, we propose a filter based on a player's average minutes per game, conditional on the player appearing in the game. This avoids penalizing strong players who missed large portions of the season and would otherwise have artificially low average minutes.

7 Grid Search

We conducted a grid search over minimum average-minutes-per-game thresholds,

$$\text{average_minutes}_i \in \{0, 5, 10, 15, 20, 25, 30\},$$

where average_minutes_i denotes the average minutes per game for player i , conditional on the player appearing in the game.

Using these new matrices which had different sizes due to the filtering of players we trained using the regular season games up until march, setting aside March and April as the test set. Running both the standard RAPM model and the Dummy RAPM with the additional 5 columns.

Table 1: Summary of RMSE and R^2 by Minimum Minutes Threshold

Min. Min.	Dummy		No Dummy	
	RMSE	R^2	RMSE	R^2
0	13.44 ± 1.17	0.100 ± 0.060	13.44 ± 1.18	0.100 ± 0.060
5	13.23 ± 1.03	0.128 ± 0.044	13.29 ± 1.05	0.119 ± 0.044
10	13.21 ± 1.03	0.130 ± 0.039	13.29 ± 1.08	0.120 ± 0.038
15	13.25 ± 1.07	0.125 ± 0.038	13.31 ± 1.11	0.117 ± 0.038
20	13.29 ± 1.09	0.120 ± 0.030	13.34 ± 1.13	0.114 ± 0.029
25	13.37 ± 1.08	0.110 ± 0.032	13.41 ± 1.10	0.105 ± 0.029
30	13.42 ± 1.14	0.104 ± 0.032	13.45 ± 1.15	0.099 ± 0.031

To evaluate out-of-sample predictive performance, both models were compared across 20 seasons of data. Figure 1 presents the R^2 values for each model across all 20 seasons, with significance levels derived from a paired t -test used to assess whether differences in mean performance were statistically meaningful.

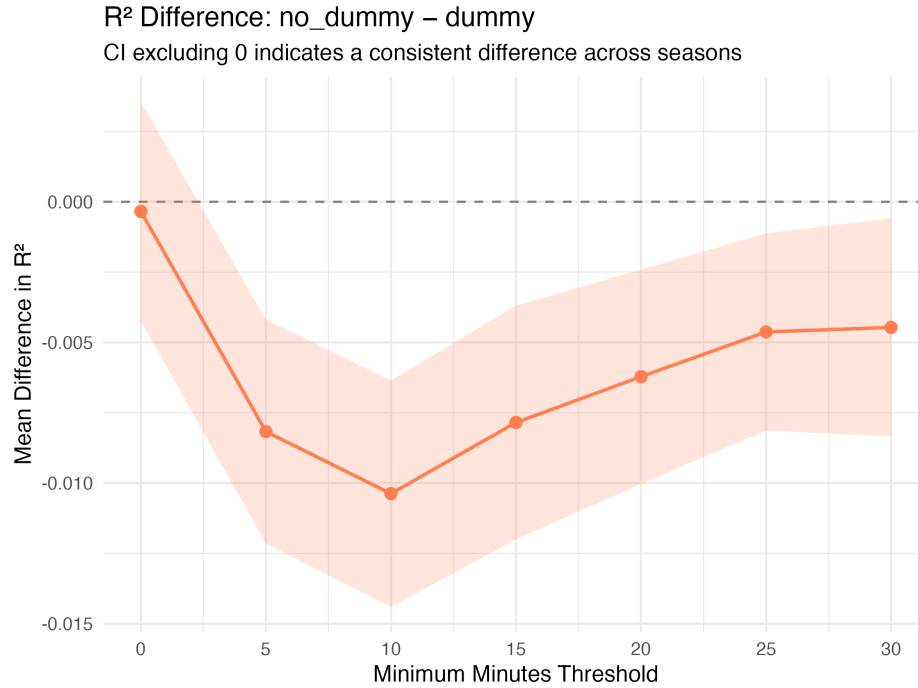


Figure 1: Out-of-sample R^2 for the dummy and no-dummy models across 20 seasons. The dummy model consistently achieves higher R^2 values, and a paired t -test confirms that the difference in mean performance between the two models is statistically significant.

8 Model Comparisons

8.1 Dummy against No Dummy at 10 Minute Filter

To assess the consistency of model performance, we compare the out-of-sample RMSE of the dummy and no-dummy models across the last 20 seasons using the optimal minimum minutes threshold of 10 minutes. The dummy model achieves a lower RMSE in 19 out of 20 seasons, suggesting a robust improvement rather than an artifact of any particular season.

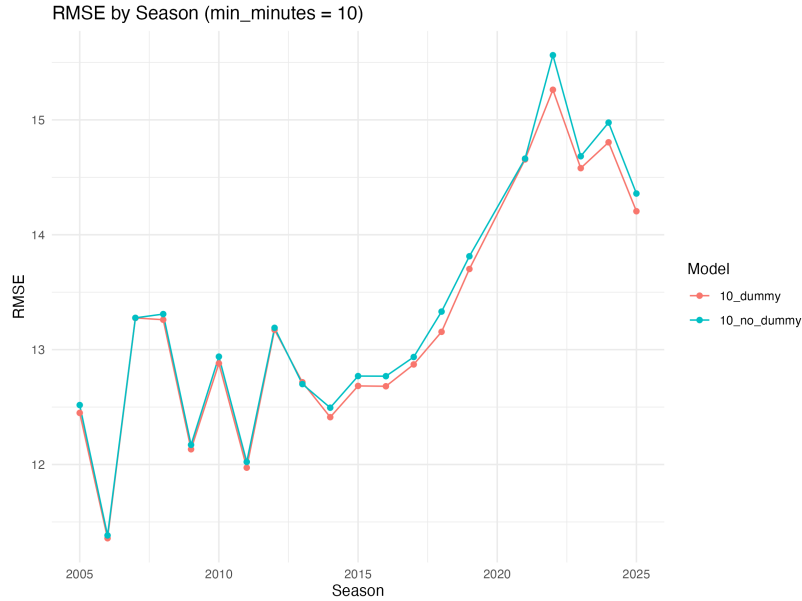


Figure 2: Out-of-sample $RMSE$ for the dummy and no-dummy models across 20 seasons using a minimum of 10 minutes for a qualified player. In all but one season do we see a lower $RMSE$ for the dummy model.

8.2 Dummy Against No Filtering

Filtering out players with fewer than 10 minutes played improved the dummy model's explanatory power over no filtering in 16 of 20 seasons, often substantially. This suggests that low-minute players introduce noise that the unfiltered model overfits to, rather than signal that generalizes; removing them allows the model to explain more of the variance in the remaining, more reliable observations.

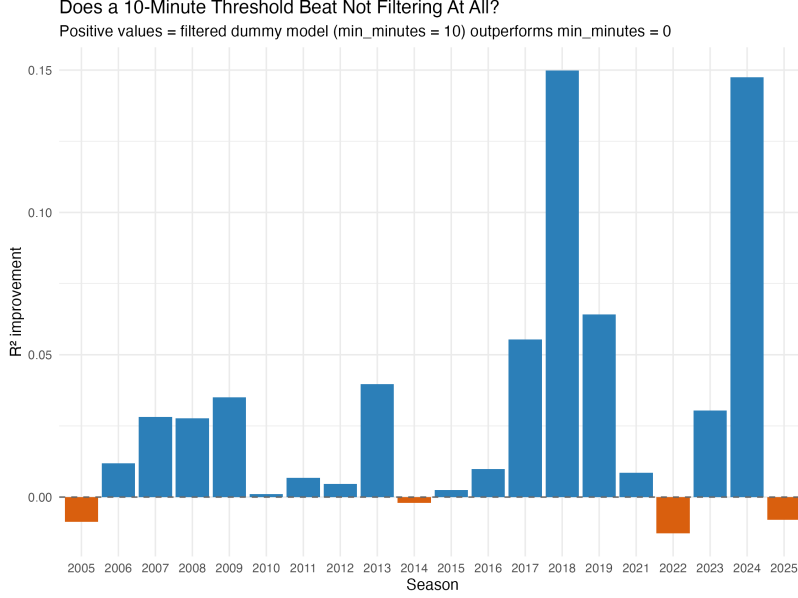


Figure 3: Out-of-sample R^2 for the 10 minute dummy model against no filtering of players. Across 20 seasons the dummy model outperforms in 16 of them.

9 Dummy Variable Coefficients

Table 2: Dummy Player Coefficients — Mean with 95% CI and One-Sided p -value ($H_0: \mu = 0$, $H_1: \mu < 0$)

Unqualified Players	Mean Coefficient (95% CI)	p -value
1	-0.447 ± 0.109	3.0×10^{-8}
2	-0.716 ± 0.239	2.6×10^{-6}
3	-1.400 ± 0.421	6.3×10^{-7}
4	-1.520 ± 1.057	0.004
5	-0.593 ± 1.825	0.252

As expected, the coefficients assigned to dummy players are negative across all levels, suggesting that lineups containing players who fall below the minimum minutes threshold are associated with a lower net point differential. The units for these coefficients is net points per 100 possessions.

For lineups with up to four unqualified players, this effect is statistically significant at the 5% level. Apart from having five unqualified players in the lineup — partially reflecting the rarity of such lineups — the coefficients become more negative as we add more unqualified players.

References

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